Solution 1 Brownian Bridge

Problem 1 — We set $W_t = B_t - tB_1$ for every $t \in [0,1]$.

Problem 1.1 — Show that $(W_t)_{t\in[0,1]}$ is a centered Gaussian process and give its covariance function.

Proof. We have that W_t is a linear combination of Gaussian processes, and so is itself Gaussian. It is centered since B_t is centered. The covariance function is given by

$$\begin{aligned}\operatorname{cov}(W_s, W_t) &= \operatorname{cov}(B_s - sB_1, B_t - tB_1)\\ &= \operatorname{cov}(B_s, B_t) - s\operatorname{cov}(B_1, B_t) - t\operatorname{cov}(B_s, B_1) + st\operatorname{cov}(B_1, B_1)\\ &= s \wedge t - st,\end{aligned}$$

where we have used the fact that B_t is a Brownian motion, so that $\operatorname{cov}(B_s, B_t) = s \wedge t$. □

Problem 1.2 — Let $0 < t_1 < t_2 < \cdots < t_p < 1$. Show that the law of $(W_{t_1}, W_{t_2}, \ldots, W_{t_p})$ has density

$$g(x_1, x_2, \ldots, x_p) = \sqrt{2\pi} p_{t_1}(x_1) p_{t_2-t_1}(x_2 - x_1) \cdots p_{t_p-t_{p-1}}(x_p - x_{p-1}) p_{1-t_p}(-x_p),$$

where $p_t(x) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t}$, and explain why the law of $(W_{t_1}, W_{t_2}, \ldots, W_{t_p})$ can be interpreted as the conditional law of $(B_{t_1}, B_{t_2}, \ldots, B_{t_p})$ knowing that $B_1 = 0$.

Proof. Observe that $B_t - tB_1$ is independent of B_1 for all $t \in [0,1]$. Indeed,

$$\begin{aligned}\operatorname{cov}(B_t - tB_1, B_1) &= \operatorname{cov}(B_t, B_1) - t\operatorname{cov}(B_1, B_1)\\ &= t - t\\ &= 0.\end{aligned}$$

Thus, the density functions of $(W_{t_1}, \ldots, W_{t_p}, B_1)$ (denoted f) and B_1 (denoted p_1) will satisfy

$$f(x_1, \ldots, x_p, y) = g(x_1, \ldots, x_p) p_1(y).$$

Idea — Everything here is a GRV with support on its domain (I can draw any small ball in $\mathbb{R}^{p+1}$ and there is some nonzero chance that $(W_{t_1}, \ldots, W_{t_p}, B_1)$ will lie in that ball, etc), so all of these densities exist.

However, it's pretty easy to compute f. Let h be the density function of $(B_{t_1}, \ldots, B_{t_p}, B_1)$.

$$f(x_1, \ldots, x_p, y) = h(x_1 + t_1 y, \ldots, x_p + t_p y, y).$$

Idea (The Jacobian is one) — This linear transformation is the matrix

$$\begin{pmatrix} 1 & t_1 & 0 & \cdots & 0 \\ 0 & 1 & t_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix},$$

which has determinant one. So when we use change of variables, there's no Jacobian to worry about.

We know h and p_1 though. Let $t_0 = 0$, $x_0 = 0$, $t_{p+1} = 1$, and $x_{p+1} = 0$.

$$h(x_1 + t_1 y, \ldots, x_p + t_p y, y) = \prod_{i=1}^{p+1} \frac{1}{\sqrt{2\pi(t_i - t_{i-1})}} e^{-((x_i + t_i y) - (x_{i-1} + t_{i-1} y))^2 / 2(t_i - t_{i-1})}$$

Just stare at the exponents for a while. We have

$$\begin{aligned} \sum_{i=1}^{p+1} \frac{((x_i + t_i y) - (x_{i-1} + t_{i-1} y))^2}{2(t_i - t_{i-1})} &= \sum_{i=1}^{p+1} \frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})} \\ &\quad + 2y \sum_{i=1}^{p+1} \frac{(x_i - x_{i-1})(t_i - t_{i-1})}{2(t_i - t_{i-1})} + y^2 \sum_{i=1}^{p+1} \frac{(t_i - t_{i-1})^2}{2(t_i - t_{i-1})} \\ &= \sum_{i=1}^{p+1} \frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})} + y^2. \end{aligned}$$

Thus,

$$h(x_1 + t_1 y, \ldots, x_p + t_p y, y) = \left(\prod_{i=1}^{p+1} \underbrace{\frac{1}{\sqrt{2\pi(t_i - t_{i-1})}} e^{-(x_i - x_{i-1})^2 / 2(t_i - t_{i-1})}}_{p_{t_i - t_{i-1}}(x_i - x_{i-1})} \right) p_1(y).$$

and

$$g(x_1, \ldots, x_p) = \sqrt{2\pi} p_{t_1}(x_1) p_{t_2 - t_1}(x_2 - x_1) \cdots p_{t_p - t_{p-1}}(x_p - x_{p-1}) p_{1 - t_p}(-x_p).$$

as desired. □

Problem 1.3 — Verify that the two processes $(W_t)_{t \in [0,1]}$ and $(W_{1-t})_{t \in [0,1]}$ have the same distribution (similarly as in the definition of Wiener measure, this law is a probability measure on the space of all continuous functions from $[0, 1]$ into $\mathbb{R}$).

Proof. The density we just showed is manifestly time-reversal-invariant. Of course, prescribing this density is equivalent to specifying the law on all finite-dimensional marginals, i.e. the probability measure over any cylinder set consisting of continuous functions which pass through Borel sets $A_1, \ldots A_n$ at times $t_1 < \cdots < t_n$. These cylinder sets form a generating π-system for the product σ-algebra, and thus the law of $(W_t)_{t \in [0,1]}$ is time-reversal-invariant. □

Solution 2 Time Reversal for Brownian Motion

Problem 2 — We set $B_0^t = B_1 B_1^t$ for every $t \in [0,1]$. Show that the two processes $(B_t)_{t\in[0,1]}$ and $(B_0^t)_{t\in[0,1]}$ have the same law (as in the definition of Wiener measure, this law is a probability measure on the space of all continuous functions from $[0,1]$ into $\mathbb{R}$).

Idea (Hint) — Note that the choice of end time $t = 1$ is not special: it will follow by essentially the same argument that for any fixed $t \geq 0$, the process $(B_t - B_{t-s} : 0 \leq s \leq t)$ is a standard Brownian motion on the time interval $[0,t]$.

Solution 3 Reflected Brownian Motion

Problem 3 — Let B_t be a standard Brownian motion, $S_t = \sup\{B_s : 0 \le s \le t\}$ its running maximum, and $Y_t = S_t - B_t \ge 0$. In this exercise you will show that the processes $(Y_t)_{t\ge 0}$ and $(|B_t|)_{t\ge 0}$ are equal in law. Note: be careful that $(S_t)_{t\ge 0}$ is in fact not a Markov process, although it will follow from this exercise that $(Y_t)_{t\ge 0}$ is.

Problem 3.1 — First argue that to prove the claim, it suffices to show that for any fixed $s, t \ge 0$, conditional on $\mathscr{F}_s$ the random variable Y_{s+t} is equidistributed as $\left|Y_s + \bar{B}_t\right|$ where $\bar{B}$ denotes a Brownian motion independent of $\mathscr{F}_s$.

What it means for $(Y_t)_{t\ge 0}$ and $(|B_t|)_{t\ge 0}$ to have the same law is that for all $t_1, t_2, \ldots, t_n \ge 0$, the joint distribution of $(Y_{t_1}, Y_{t_2}, \ldots, Y_{t_n})$ is the same as the joint distribution of $(|B_{t_1}|, |B_{t_2}|, \ldots, |B_{t_n}|)$. The key is to break this up into conditional distributions, and then use the fact that the law of $(B_t)_{t\ge 0}$ is invariant under time reversal.

Start with the expression

$$\mathbb{P}\left(Y_{s+t} \in A \mid \mathscr{F}_s\right) = \mathbb{P}\left(\left|Y_s + \bar{B}_t\right| \in A \mid \mathscr{F}_s\right), \quad a \ge 0, \qquad A \in \mathscr{B}(\mathbb{R}).$$

We will now show by induction that

$$\mathbb{P}(\bigcap_i (Y_{t_i} \in A_i)) = \mathbb{P}(\bigcap_i (|B_{t_i}| \in A_i))$$

for all $t_1, t_2, \ldots, t_n \ge 0$ and $A_1, A_2, \ldots, A_n \in \mathscr{B}(\mathbb{R})$. The base case is $n = 0$ which is trivial.

Then, as the inductive step,

$$\begin{aligned}
\mathbb{P}\left(\bigcap_i (Y_{t_i} \in A_i)\right) &= \mathbb{P}\left(\left(\left|Y_{t_{i-1}+\bar{B}_{t_i - t_{i-1}}}\right| \in A_i\right) \cap \bigcap_{j<i} (Y_{t_j} \in A_j)\right) \\
&= \mathbb{P}\left(\left(\left|B_{t_{i-1}+\bar{B}_{t_i - t_{i-1}}}\right| \in A_i\right) \cap \bigcap_{j<i} (B_{t_j} \in A_j)\right) \\
&= \mathbb{P}\left(\bigcap_i (|B_{t_i}| \in A_i)\right).
\end{aligned}$$

where the last line is because conditioned on $\{B_{t_j} : j < i\}$, $B_{t_{i-1}} + \bar{B}_{t_i - t_{i-1}}$ has the same law as $B_{t_{i-1}+B_{t_i - t_{i-1}}} = B_{t_i}$.

The inductive hypothesis is precisely the claim that $(Y_t)_{t\ge 0}$ and $(|B_t|)_{t\ge 0}$ have the same law (we have verified it over all finite-dimensional cylinder sets, which form a π-system generating the product σ-algebra). Thus, we have shown that the two processes have the same law.

Problem 3.2 — For $u \ge 0$ let $\hat{B}_u \equiv B_{s+u} - B_s$ and $\hat{S}_u = \sup\{\hat{B}_w : 0 \le w \le u\}$. Show that $Y_{s+t} = \max\{Y_s, \hat{S}_t\} - \hat{B}_t$.

Problem 3.3 — Argue that for any fixed $y \geq 0$, $\max\{y, \hat{S}_t\} - \hat{B}_t$ has the same law as $\left|y + \hat{B}_t\right|$.

Idea (Hint) — A hint is to first decompose

$$\mathbb{P}\left(\max\{y, \hat{S}_t\} - \hat{B}_t > a\right) = \mathbb{P}\left(y - \hat{B}_t > a\right) + \mathbb{P}\left(y - \hat{B}_t \leq a, \hat{S}_t - \hat{B}_t > a\right),$$

and use time reversal to simplify the last term.

Solution 4 Arcsine Law

Problem 4 — Set $T = \inf\{t \geq 0 : B_t = S_1\}$.

Problem 4.1 — Show that $T < 1$ a.s. (one may use the result of the previous exercise) and then that T is not a stopping time.

We know the Brownian motion is continuous almost surely. Continuous functions on compact domains attain their maximum, thus there exists a $t \in [0, 1]$ such that $B_t = S_1$ almost surely. It remains to show that $B_1 = S_1$ almost never. To do this, we use time reversal; let $B'_t = B_{1-t} - B_1$, and note that B'_t is a Brownian motion; then $S'_1 = \inf\{B'_t, 0 \leq t \leq 1\} = S_1 - B_1$. However, as a Brownian motion, we know that almost surely for every $\varepsilon > 0$,

$$\sup_{0 \leq t \leq \varepsilon} B'_t > 0$$

thus $S'_1 > 0$ almost surely, i.e. $S_1 > B_1$ almost surely. So we conclude.

Problem 4.2 — Verify that the three variables S_t, $S_t - B_t$, and $|B_t|$ have the same law.

Problem 4.3 — Show that T is distributed according to the so-called arcsine law, whose density is

$$g(t) = \frac{1}{\pi\sqrt{t(1-t)}} \mathbb{1}_{(0,1)}(t).$$

Problem 4.4 — Show that the results of questions 1. and 3. remain valid if T is replaced by $L = \sup\{t \leq 1 : B_t = 0\}$.

Idea (Hint) — You may use the result of Problem 3 in your solution; it is especially useful for the last part of this exercise.

Solution 5 Law of the Iterated Logarithm

Problem 5 — The goal of the exercise is to prove that

$$\limsup_{t\to\infty} \frac{B_t}{\sqrt{2t\log\log t}} = 1 \quad \text{a.s.}$$

We set $h(t) = \sqrt{2t\log\log t}$.

Idea (Hint) — In your solution, please give a proof of the following standard Gaussian tail bound: if $Z \sim \mathcal{N}(0,1)$, then for all $x > 0$ we have

$$\left(1 - \frac{1}{x^2}\right)\exp\left(-\frac{x^2}{2}\right) \le \mathbb{P}(Z \ge x) \le \exp\left(-\frac{x^2}{2}\right)\frac{1}{x\sqrt{2\pi}}.$$

Problem 5.1 — Show that, for every $t > 0$, $\mathbb{P}(S_t > u\sqrt{t}) \sim \frac{2}{u\sqrt{2\pi}}e^{-u^2/2}$ when $u \to +\infty$.

Problem 5.2 — Let r and c be two real numbers such that $1 < r < c^2$. From the behavior of the probabilities $\mathbb{P}(S_{r^n} > ch(r^{n-1}))$ when $n \to 1$, infer that, a.s.,

$$\limsup_{t\to\infty} \frac{B_t}{h(t)} \ge 1.$$

Problem 5.3 — Show that a.s. there are infinitely many values of n such that

$$B_{r^n} - B_{r^{n-1}} \ge \sqrt{\frac{r-1}{r}}h(r^{n-1}).$$

Conclude that the statement given at the beginning of the exercise holds.

Problem 5.4 — What is the value of $\liminf_{t\to\infty} \frac{B_t}{h(t)}$?